way to guarantee that you won't get screwed by your portfolio manager. But the situation is not all dismal. I believe many useful principles are available to asset owners, but some of these unfortunately are neglected in industry practice.

3.2. IRRELEVANCE RESULT

A common way to reward a fund manager is to pay her for beating a benchmark, which is typically set to be an asset class index like the S&P 500. The agent's compensation is typically structured in a *linear* way, so she gets a base salary and then some fraction or multiple of the outperformance. We often choose linear contracts not because they are optimal (they generally aren't), but because they are convenient. In standard agency problems, linear contracts satisfy the participation constraint (a high enough base salary entices the talented agents to come to work) and the incentive-compatibility constraint (as the talented agent has an incentive to work hard).

Amazingly, the linear contract does not get the manager to behave in an optimal way for the asset owner. This *Irrelevance Result*, discovered by Stoughton (1993) and Admati and Pfleiderer (1997), states that *linear contracts have no useful role in delegated portfolio management*. The agent, who controls both the mean and risk of the portfolio, can always undo what the principal wants to achieve. Put another way, the "Irrelevance Result" is that the manager's effort does not depend on how much he gets paid. This is remarkable because linear payments for outperformance of asset class indices are so common, yet they are not beneficial to asset owners.

The setup of Stoughton-Admati-Pfleiderer is very realistic: the principal selects from multiple agents, just like the asset owner has a multiplicity of (supposedly talented) fund managers to choose from, all hawking their own brand of alpha, and there is hidden effort which arises because asset owners cannot look over the shoulders of all the fund managers they hire to determine whether or not they are shirking. The linear benchmarks based on the S&P 500, or similar asset class benchmarks, do not optimally share risk, do not allow the principal to achieve optimality, weaken a manager's incentives to expend effort, are not useful in screening out bad managers, and play no role in aligning manager preferences with the investor's.

The Irrelevance Result highlights the importance of the optimal portfolio of the asset owner. We cannot talk about the agency relation between the fund manager and the asset owner without first determining the optimal portfolio for the asset owner. In chapter 14, we saw that this must be based on a foundation of factor risk. Thus there are two decisions that cannot be delegated to agents: the level of risk to be taken and the key sources of risk premiums to be exploited, which both depend on the characteristics of the investor and her liabilities, income, and wealth (see Parts I and II of this book). Getting this benchmark right is step one so that all agency distortions can be measured relative to it.